



Stable metal-organic frameworks based mixed matrix membranes for Ethylbenzene/N₂ separation

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ABSTRACT

To remove volatile organic compounds (VOCs) from air, N₂ or other waste gas streams in industrial processes is significant to the environmental protection and human health, as well as gain the opportunity to reuse these valuable chemicals. Mixed matrix membranes (MMMs) based on metal-organic frameworks (MOFs) have been proved to have potential application in this field. In this work, MIL-101 and UiO-66 are selected to construct MMMs for efficient ethylbenzene separation because of following factors: (1) the high ethylbenzene adsorption capacities; (2) the three-dimensional channels with large pore size; and (3) the ideal hydrothermal stability of structures. MIL-101@Pebax and UiO-66@Pebax MMMs have been prepared in an environment-friendly way and evaluated for the ethylbenzene/N₂ separation performance. SEM, TGA, FTIR and XRD results indicate that continuous MMMs with different filler loading ratios have been fabricated successfully. The results show that MIL-101@Pebax and UiO-66@Pebax with the MOF loading ratio of 20 wt% possess ethylbenzene/N₂ permselectivity of 284 and 100 respectively at 25 °C, 0.05 MPa and moderate feed ethylbenzene concentration (1000 ppm), which increased by 10.5 and 3.0 times compared with the pristine Pebax membrane. Effects of upstream pressure, feed concentration, and operating temperature on separation performance have been investigated for ethylbenzene/N₂ separation. Due to the selective adsorption capacity to ethylbenzene, as well as the large pore size of MIL-101(Cr), MIL-101@Pebax can be applied as efficient membrane materials in ethylbenzene capture system for high concentration region at ambient temperature.

1. Introduction

Volatile organic compounds (VOCs) are the general term for organic chemicals being easily vaporized at room temperature and ambient pressure, which exhibit saturated vapor pressure greater than 133.32 Pa at normal atmospheric temperature. Their boiling points and the initial boiling points are usually below or equal to 250 °C. VOCs emissions into atmosphere boost the formation of tropospheric ozone, photochemical smog, organic acids and secondary organic aerosols in the presence of sunlight, and that would cause skin and respiratory tract irritation, genetic modification, or even cancer of mammals, so do the human. Ethylbenzene is volatile and it can enter the human body easily through respiratory tract, mucosa and even skin [1], possessing toxic effects on

human nervous, digestive, immune and reproductive system [2]. Ethylbenzene has been included in the class 2B in the list of carcinogens published by International Agency for Research on Cancer of World Health Organization [3,4]. Ethylbenzene leakage usually occurs in petrochemical plants such as ethylbenzene dehydrogenation unit and ethylbenzene-styrene plants, as well as in the storage and transportation process such as tank breather valves and pipeline sealing points, it's extremely harmful to the health of working personnel and residents nearby [5]. As people pay more attention to the health hazards associated with VOCs, the research on the abatement of VOCs has been attracting continued attention. So far, most of the existing technologies for controlling organic vapor emissions, such as biological treatment [6–8], adsorption [9–11], absorption [12–14] and condensation

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